

POLARIS

Antarctic Iceberg Trajectory & Navigation Decision Support System

Project Development Roadmap

Based on the supplied SRD. The core workflow is satellite data acquisition, processing, iceberg/sea-ice detection, AI/ML trajectory prediction, risk assessment and navigation decision support.

Phase 1 — Project Foundation

- Define Polaris architecture
- Set up Python environment
- Install GDAL, Rasterio, NumPy, OpenCV and AI/ML tools
- Create repository and modular project structure
- Define data formats and storage strategy

Phase 2 — NASA MODIS Data Acquisition

- Connect to NASA Earthdata / LAADS DAAC
- Select Antarctic geographic region
- Add date/time based retrieval
- Download MODIS observations and metadata
- Validate downloaded datasets

Phase 3 — Satellite Data Processing

- Read satellite raster data
- Crop Antarctic region
- Reproject and resample data
- Extract required bands and metadata
- Generate clean processed datasets

Phase 4 — Iceberg & Sea-Ice Detection

- Enhance satellite imagery
- Reduce noise and segment images
- Detect ice regions using computer vision
- Develop and evaluate an AI/ML detection model
- Extract iceberg boundaries and positions

Phase 5 — Feature Extraction

- Latitude and longitude
- Iceberg position and boundary
- Ice-covered area
- Distance from ship
- Reflectance, brightness and spectral information
- Sea-surface temperature and cloud information

- Observation timestamps and historical movement

Phase 6 — Iceberg Tracking

- Compare observations over time
- Calculate movement distance and direction
- Estimate movement velocity
- Build historical iceberg trajectories

Phase 7 — AI/ML Trajectory Prediction

- Create a simple prediction baseline
- Train ML/time-series models using historical features
- Predict future iceberg position
- Generate predicted trajectory
- Measure prediction error

Phase 8 — Risk Assessment

- Combine ship position with predicted iceberg trajectory
- Calculate separation/closest approach
- Estimate risk level
- Classify situations as Low, Medium or High Risk

Phase 9 — Navigation Decision Support

- Receive ship position from GPS
- Combine detection, prediction and risk outputs
- Calculate safer route recommendations
- Prepare navigation-support output for ECDIS
- Keep navigation decisions separate from the MODIS data-source layer

Phase 10 — Offline Mode

- Store required processed observations and features
- Run local processing without connectivity
- Use previously downloaded satellite data
- Synchronize newly available data when connectivity returns

Phase 11 — Polaris Dashboard

- Antarctic map interface
- Ship position
- Iceberg locations and boundaries
- Predicted iceberg paths
- Risk indicators
- Recommended safer route

Phase 12 — Testing & Validation

- Validate satellite data quality
- Evaluate iceberg detection accuracy

- Compare predicted vs actual positions
- Test risk scenarios
- Test online/offline operation
- End-to-end system testing

POLARIS — Overall System Flow

Step	Component	Purpose
1	NASA Aqua / MODIS	Satellite observation
2	Earthdata / LAADS DAAC	Data search and retrieval
3	Data Validation	Verify downloaded observations
4	GDAL / Rasterio	Raster and geospatial processing
5	NumPy / OpenCV	Image enhancement and feature extraction
6	Ice Detection	Sea-ice / iceberg identification
7	AI/ML	Trajectory prediction
8	Risk Assessment	Evaluate ship–iceberg risk
9	GPS + Ship Position	Current vessel location
10	Navigation Support	Recommend safer route
11	ECDIS	Display navigation decision-support output

Recommended MVP Sequence

1. Download one Antarctic MODIS observation.
2. Process and crop the required region.
3. Detect ice/iceberg regions.
4. Extract iceberg coordinates and features.
5. Track movement across multiple observations.
6. Predict the next iceberg position.
7. Add ship position.
8. Calculate risk / closest approach.
9. Display a safer-route recommendation on a map.

SRD alignment: The supplied SRD defines MODIS as the satellite observation/data-source layer and specifies the chain through AI/ML, iceberg trajectory, risk assessment and navigation recommendation. It also specifies offline processing, storage efficiency, reliability, scalability and modular maintainability.